

ML4HOR: 9 Stages and 28 Recommendations

Manuscript-preparation checklist

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Purpose

This document exposes the complete **ML4HOR framework** as a compact checklist: **9 stages** and **28 numbered recommendations** for designing, analysing, reporting, and translating machine-learning studies in health outcomes research.

Recommendations map

Stage	Domain	Recommendation numbers
1	Clinical problem formulation	R1-R3
2	Data foundations	R4-R7
3	Feature engineering	R8-R10
4	Model development	R11-R14
5	Model validation	R15-R17
6	Performance assessment	R18-R20
7	Interpretability	R21-R23
8	Reporting package	R24-R26
9	Clinical translation	R27-R28

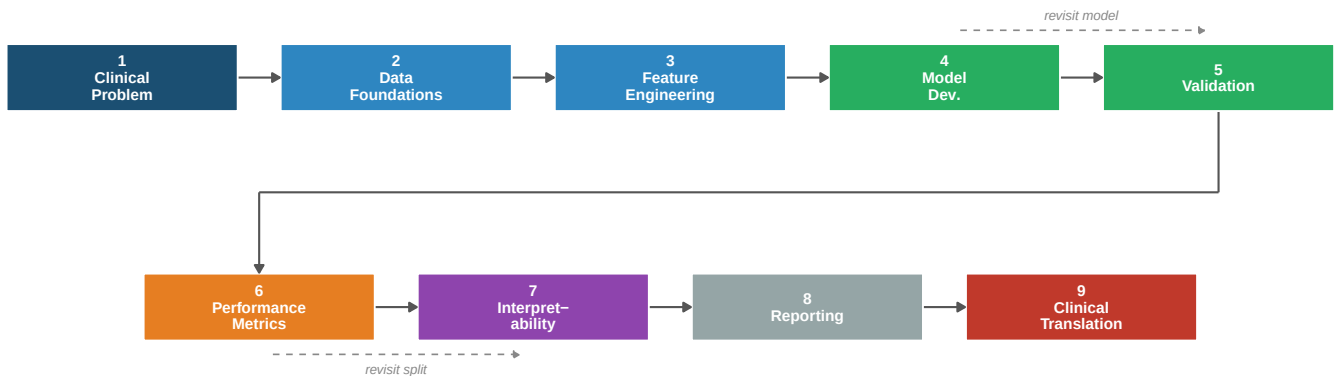


Figure 1

Full checklist

Stage 1. Clinical Problem Formulation

R1. Formalise the clinical prediction task as a PICO-ML specification in writing before any data access. Define the target population, index prediction task, comparator, outcome operationalisation, data modality, and ML task class.

R2. Select the ML task formulation based strictly on outcome type; do not dichotomise a survival outcome at an arbitrary time horizon for analytical convenience.

R3. Pre-specify the primary performance metric and the clinical decision threshold *a priori*; register both in a pre-analysis plan or data access agreement.

Stage 2. Data Foundations

R4. Conduct a structured data quality audit covering completeness, accuracy, consistency, timeliness, and provenance before modelling; document findings as supplementary material.

R5. Assess missing data mechanisms for each key predictor; apply multiple imputation under MAR, crediting Donald B. Rubin's foundational framework; conduct sensitivity analyses for MNAR predictors; never use single mean imputation as the primary strategy.

R6. Use a temporal split aligned with the deployment context for any EHR, registry, or administrative data with temporal ordering; reserve geographic splits to explicitly evaluate transportability.

R7. Determine minimum sample size *a priori* using formal methods; report the class distribution and address outcome prevalence below 10% with imbalance-aware methods and AUPRC as the primary discrimination metric.

Stage 3. Feature Engineering

R8. Construct a temporal data flow diagram for all candidate features; exclude any variable not reliably available at the prediction time point.

R9. Model continuous predictors with restricted cubic splines using 3–5 knots unless the model is intended for point-of-care scoring with manual feature entry.

R10. Document all preprocessing decisions, including encoding, scaling, imputation sequence, and feature exclusion criteria, in reproducible, version-controlled code.

Stage 4. Model Development

R11. Fit a simple interpretable baseline, such as logistic regression for classification or Cox PH for survival, before complex alternatives; justify added complexity by a performance gain exceeding the 95% CI of the baseline.

R12. Choose the loss function to match the target performance metric; do not optimise accuracy when discrimination, calibration, or clinical utility is the scientific objective.

R13. Tune hyperparameters on a held-out validation set or the inner loop of nested cross-validation; never select the final model based on test-set performance.

R14. For censored survival outcomes, use survival-specific algorithms such as Cox proportional hazards models, random survival forests, or other censoring-aware learners; document the handling of tied event times and administrative censoring.

Stage 5. Model Validation

R15. Report discrimination and calibration metrics in at least two pre-specified demographic subgroups; report equalised-odds and calibration-across-groups metrics for equity evaluation.

R16. Perform temporal validation before deployment; pre-specify performance-degradation triggers, such as AUC below a threshold or calibration slope outside a defined interval, that mandate retraining.

R17. For multi-site deployment, obtain performance estimates at a minimum of one geographically distinct external site before generalised clinical use.

Stage 6. Performance Assessment

R18. Report discrimination, calibration, and clinical utility for every model; plot calibration curves at predicted probability deciles; plot decision curve analysis at the clinically relevant threshold range.

R19. For outcomes with prevalence below 10%, report AUPRC as the primary discrimination metric; supplement with Brier score.

R20. Recalibrate miscalibrated models, such as calibration slope below 0.8 or above 1.2, using Platt scaling or isotonic regression before clinical deployment; report post-recalibration performance.

Stage 7. Interpretability

R21. Apply at least one global interpretability method, such as permutation importance or SHAP, and one local method, such as SHAP waterfall or LIME, to every non-linear model before clinical use.

R22. Have a clinical domain expert validate the direction and magnitude of top-ranked SHAP features against established pathophysiology; document any clinically implausible associations.

R23. Do not present attention weights as causal explanations; validate post-hoc explanations against internal consistency checks such as feature ablation or perturbation tests.

Stage 8. Reporting Package

R24. Complete and submit the TRIPOD+AI checklist with every prediction model manuscript.

R25. Complete a PROBAST+AI assessment of methodological quality, risk of bias, and applicability; address identified concerns in the manuscript or supplementary materials.

R26. Deposit all analysis code in a public repository; share data or a synthetic equivalent; publish a model card before clinical deployment.

Stage 9. Clinical Translation

R27. Plan the clinical evaluation pathway explicitly using DECIDE-AI before live deployment; prospective clinical evaluation must precede policy adoption.

R28. Establish pre-specified drift monitoring thresholds and a retraining protocol before deployment; publish a model card documenting intended use, failure modes, demographic scope, and post-deployment performance tracking plan.

Core references

1. Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3), 581-592.
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3. Collins, G. S., Moons, K. G., Dhiman, P., Riley, R. D., Beam, A. L., Van Calster, B., ... & Logullo, P. (2024). TRIPOD+ AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *bmj*, 385.
4. Moons, K. G., Damen, J. A., Kaul, T., Hooft, L., Navarro, C. A., Dhiman, P., ... & van Smeden, M. (2025). PROBAST+ AI: an updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods. *bmj*, 388.
5. Vasey, B., Nagendran, M., Campbell, B., Clifton, D. A., Collins, G. S., Denaxas, S., ... & McCulloch, P. (2022). Reporting guideline for the early stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. *bmj*, 377.